

COURSE CODE: MLA03 **COURSE NAME:** Reinforcement Learning

COURSE OUTCOME: CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)

FACULTY: Dr. G. A. SENTHIL **TOTAL MARKS:** 50 Marks (40% Weightage) Duration: 4 Hrs

Q1. Flowchart of Model-Based RL Workflow (Environment Interaction to Policy Optimization) [5 Marks]

Workflow Explanation: Model-Based Reinforcement Learning systematically alternates between real-world exploration, parameter estimation of the transition dynamics & reward functions, trajectory rollouts (simulated imagination), and policy/value improvement.



- **Step 1: Real-World Rollout:** Agent interacts with environment using current policy π_θ and stores transition tuples (s_t, a_t, r_t, s_{t+1}) into replay buffer D .
- **Step 2: Supervised Model Fitting:** Learn transition dynamics model f_ϕ by minimizing regression loss: $L(\phi) = E_{\{(s,a,s') \sim D\}} [\|f_\phi(s,a) - s'\|^2]$.
- **Step 3: Simulation & Optimization:** Generate virtual trajectories via f_ϕ and update policy parameters θ using either sampling-based trajectory planning or analytic policy gradients.

Q2. Concept Map: Model-Based RL vs. Model-Free RL Comparison [5 Marks]

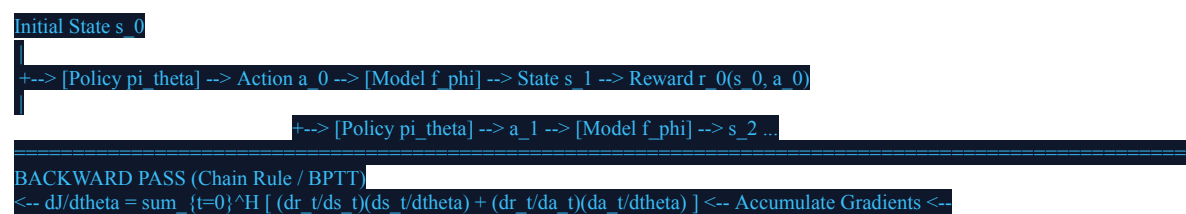
Evaluation Dimension Model-Free RL (e.g., PPO, SAC, DQN) Model-Based RL (e.g., MBPO, PETS, MuZero)

Environment Knowledge Direct trial-and-error; learns value/policy directly without explicit state transition dynamics.

Sample Efficiency Low (requires millions of real environment interactions to converge).

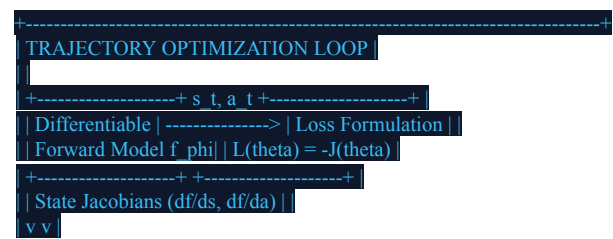
Computational Complexity	Lower compute per interaction step; fast policy updates.	High (orders of magnitude fewer real samples; leverages virtual rollouts/imagination).
Asymptotic Bias & Limits	Unbiased convergence to optimal policy; superior asymptotic performance.	High compute overhead due to iterative model fitting, trajectory forecasting, and planning. Prone to model exploitation and compounding prediction error bias over long horizons.
	Explicitly models state transitions $s_{t+1} \sim P_{\hat{}}(s' s,a)$ and rewards $R_{\hat{}}(s,a)$.	

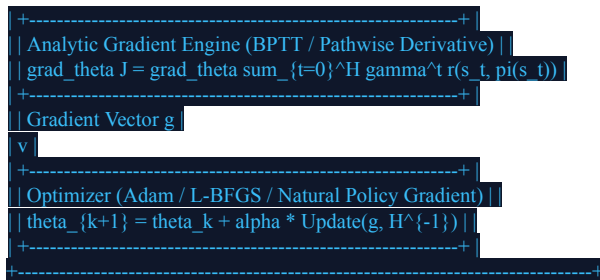
Q3. Flowchart: Analytic Gradient Computation & Backpropagation Through Dynamics [5 Marks] Core Mechanism: When both the transition dynamics model $s_{t+1} = f_{\phi}(s_t, a_t)$ and policy $a_t = \pi_{\theta}(s_t)$ are parameterized by differentiable neural networks, the gradient of the total expected return $\nabla_{\theta} J(\theta)$ can be computed analytically by unrolling the computation graph over horizon H and applying Backpropagation Through Time (BPTT).



- **Forward Trajectory Pass:** Starting from s_0 , roll out deterministic path: $a_t = \pi_{\theta}(s_t)$, $s_{t+1} = f_{\phi}(s_t, a_t)$, $r_t = r(s_t, a_t)$ for $t = 0 \dots H-1$.
- **Recursive Chain Rule:** The state Jacobian propagates backward recursively: $ds_{t+1}/d\theta = (\partial f/\partial s_t)(ds_t/d\theta) + (\partial f/\partial a_t)[(\partial \pi/\partial s_t)(ds_t/d\theta) + \partial \pi/\partial \theta]$.
- **Parameter Update:** $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$ using first-order (Adam) or quasi-Newton gradient ascent.

Q4. Block Diagram: Policy Parameter Updates, Gradients, and Reward Optimization [5 Marks]





Mitigation of Exploding/Vanishing Gradients: Analytic gradients over long horizons suffer from chaotic and unstable dynamics. Modern approaches resolve this using trajectory collocation, short truncated rollouts, or regularized value baselines.

Q5. Flowchart: Sampling-Based Planning Process (Model Predictive Control & CEM) [5 Marks] Overview:

Sampling-based planners evaluate action sequences in simulation without requiring differentiable transitions, making them highly robust to discontinuities and non-convex environments.



Q6. Concept Map: Exhaustive Search Planning vs. Sampling-Based Planning [5 Marks]

Dimension Exhaustive Search (e.g., Minimax, BFS/DFS)

Search Space Coverage Evaluates all $|A|^H$ possible branches across the complete decision tree.

Action Space Compatibility Strictly limited to small discrete action spaces; intractable for continuous control.

Curse of Dimensionality Suffers exponential computational explosion:

$$O(|A|^H).$$

Optimality Guarantees Guarantees exact globally optimal path across evaluated horizon H .

Sampling-Based Planning (e.g., MPC, CEM, MCTS)

Samples a stochastic subset $K \ll |A|^H$ of high probability candidate paths.

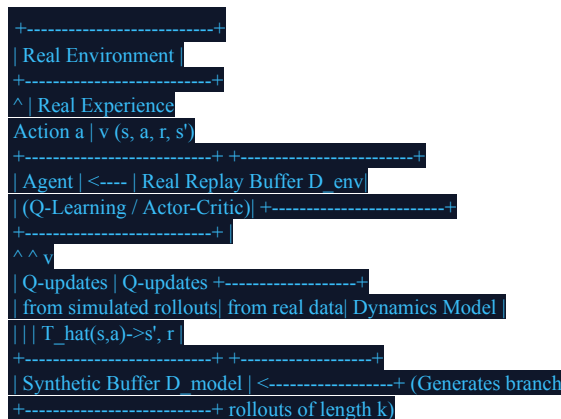
Natively handles continuous, high-dimensional control spaces via Gaussian distributions.

Scales polynomial/linearly with sample size: $O(K * H * T_{\text{eval}})$.

Asymptotically optimal or bounded sub-optimal depending on sample density K .

Q7. Block Diagram & Flowchart: Dyna-Architecture & Synthetic Data Generation [5 Marks] Dyna

Framework (Sutton): Unifies direct reinforcement learning (learning directly from real environment transitions) and indirect model-based learning (generating synthetic experience replay via the learned dynamics model).



- **Model Training:** Learns transition model T_{hat} on real dataset D_{env} .
- **Branched Rollouts:** Starts from real sampled states $s \sim D_{\text{env}}$, rolls out k -step simulated steps using policy π , and caches them in D_{model} .
- **Policy Update:** Samples mini-batches with mixing ratio $\beta D_{\text{env}} + (1-\beta) D_{\text{model}}$ to update Q-functions or Actor networks.

Q8. Concept Map: Advantages and Limitations of Model-Generated Training Data [5 Marks]

Category Key Mechanisms Impact on RL Performance

Advantages

- Sample Efficiency: Multiplies data volume 10x-100x.
- Safe Exploration: Tests risky actions in simulation.
- High Parallelism: GPU-batched vectorized rollouts.

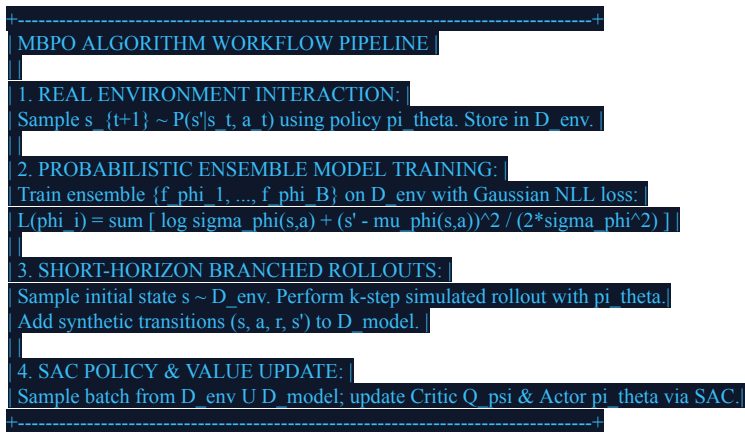
Limitations & Pitfalls

- Compounding Model Errors: Errors accumulate exponentially over horizon H .
- Model Exploitation: Policy exploits inaccurate high-reward regions.
- Distribution Shift: Policy visits out-of-distribution states.

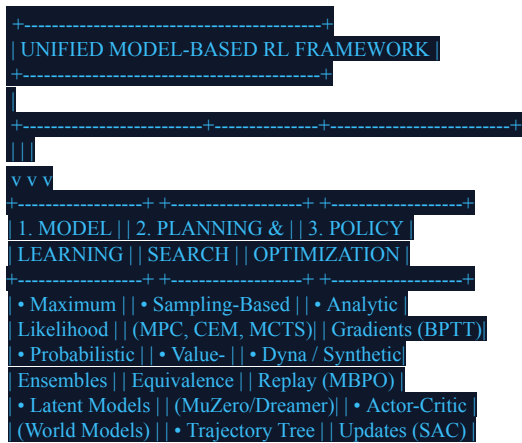
Dramatically reduces costly and hazardous real world hardware interactions. (k-step) and probabilistic ensembles.

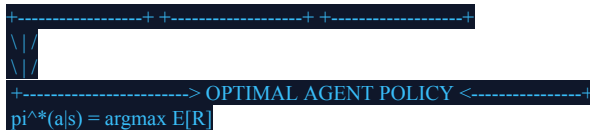
Can lead to severe policy degradation unless bounded by short rollouts

Q9. Block Diagram: Model-Based Policy Optimization (MBPO) Framework [5 Marks] MBPO Framework (Janner et al.): Guarantees monotonic policy improvement by combining bootstrapped probabilistic neural network ensembles (to quantify epistemic uncertainty) with short k-step branched rollouts initialized from real environment states.



Q10. Comprehensive Concept Map: Unified Model-Based RL Framework [5 Marks]





Synthesis: This unified architecture illustrates how model learning, value-equivalent latent dynamics, and policy optimization integrate to overcome sample inefficiency and compounding error in complex decision-making tasks.

Code of Conduct:

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Student Name: K.V. S. S. Jeswanth Babu Register Number: 192472012 Signature of the Student:

K. V. S. S. Jeswanth Babu Date: 25-08-2026

RUBRICS FOR CALCULATION

	4 –	Level 3 – Level 2 –	Level 1 –
Technical Content Accuracy	insightful technical explanation.	maps.	
Problem Identification & Formulations	25% Rigorously formulated algorithms with mathematical clarity and design flow.	15% Structured, professional, concise, and logically sequenced.	Organized and clear technical communication.
Use of Tools & Visuals	20% Advanced block diagrams,	10% Complete adherence to academic integrity and rubric standards. Technically correct content with adequate depth and clarity.	Understandable but lacks clean structure.
Communication & Organization		Clearly defined approach with logical engineering flow.	Professional presentation and adherence. Basic concepts presented with limited accuracy.
Professionalism & Integrity	30% Highly accurate, in-depth, and	Clear and effective diagrams used.	Methodology or algorithmic steps are weak.
	flowcharts, and structured concept		Unclear or incorrect statements.
			Minimal or uninformative visuals.

Poor organization;
disjointed sections.

Lacks professional
formatting.